

Unit 1

JDK Development Kit (JDK) is a software development environment which is used to develop Java applications.

JRE is an installation package that provides an environment to only run the java program(or application) onto our machine.

JVM (inbuilt in both) is responsible for executing the java program line by line, hence it is also known as an interpreter.

A **Wrapper class** is a class whose object contains primitive data types.

The automatic conversion of primitive into an object is known as **autoboxing** and vice versa unboxing.

Array is a collection of similar type of elements that have contiguous memory location.

In java, String objects are immutable which means a constant and cannot be changed once created.

There are two ways to create String object: *****By string literal *****By new keyword

StringBuffer is mutable means one can change the value of the object.

StringBuffer is synchronized i.e. thread safe. It means two threads can't call the methods of StringBuffer simultaneously.

StringBuffer is less efficient than StringBuilder.

StringBuilder is non synchronized i.e. not thread safe. It means two threads can call the methods of StringBuilder simultaneously.

StringBuilder is more efficient than StringBuffer.

super keyword can be used to access the immediate parent class constructor.

Unit 2

Java collection framework provides set of interfaces and classes to implement various data structure and algorithms

ArrayList class uses a concept of *dynamic array* for storing elements. It can contain duplicate elements also.

Method: Add(); Size(); Set(); isEmpty(); contains(); remove(); removeAll(); Indexof();

Iterator interface can traverse elements in a collection only in **forward direction**.

Method: hasNext(); next(); remove(); forEachRemaining();

ListIterator interface extends **iterator interface** and it can traverse the elements of the collection in both **forward** and **backwards** directions.

Method: hasNext(); next(); nextIndex(); previous(); previousIndex(); remove(); set();

LinkedList class uses a doubly linked list to store the elements.

Methods: add(); addFirst(); addLast(); removeFirst(); removeLast();

TreeSet class implements the Set interface that uses a tree for storage. The objects of the TreeSet class are stored in ascending order and it contains unique elements and doesn't allow null element .

Method: add(obj o); addAll(collection c); clear(); contain(obj o); First(); Last(); Size();

PriorityQueue is a queue that follow FIFO algo and it is used when objects are supposed to be processed based on the priority.

Methods: Boolean add(); remove(); public peek(); public poll();

Stack Class is a subclass of Vector that implements a standard LIFO stack. It also implements interface List, Collection framework, Iterable interface.

Methods: peek(); pop(); push(); empty(); add(obj o); add(index, Object obj); addAll(collection c); addElement(obj o)

Comparable interface

- 1) Comparable provides a **single sorting sequence**. In other words, we can sort the collection on the basis of a single element such as id, name, and price.
- 2) Comparable provides **compareTo()** method to sort elements.
- 3) Comparable is present in **java.lang** package.

Comparator interface

The Comparator provides **multiple sorting sequences**. In other words, we can sort the collection on the basis of multiple elements such as id, name, and price etc.
Comparator provides **compare()** method to sort elements.
A Comparator is present in the **java.util** package.

Properties class object contains key and value pair both as a string.

An Advantage of the properties file

Recompilation is not required if the information is changed from a properties file.

Lambda expression is just an anonymous function, i.e., a function with no name. It is used to provide the implementation of a functional interface.

An interface which has only one abstract method is called **functional interface**.

Unit 3

Multithreading in java is a process of executing multiple threads simultaneously. It is used to achieve multitasking.

Thread is a single sequential flow of execution of task of a process. They are independent, so it doesn't affect other threads if an exception occurs in a single thread.

Creating threads in Java

- Create a class that extends the Thread class
- Create a class that implements the Runnable interface

The life cycle of the thread in java is controlled by JVM. The java thread states are as follows:

—**New Born**: When we create a thread it will be in Newborn State. The thread is just created still it's not running.

—**Runnable/Running**: A thread might actually be running or it might be ready to run at any instant of time.
A thread that is ready to run is moved to **runnable state**.

In **running state** thread is in its execution mode

—**Non-Runnable/Blocked**: A thread is called in Blocked State when it is not allowed to entering in Runnable or Running State.

—**Terminated/Dead**: When a thread is completed executing its run() method the life cycle of that particular thread is end.

Thread Method: start(), run(), sleep(), join(); yield();

In multithread program where multiple threads try to access the same resources then problem occurs
To overcome this problem, we use the concept of thread synchronization.

3 methods which makes thread communication possible:

wait()-It tells the calling thread to release the lock and go to sleep until some other thread enters the same monitor and calls notify().

notify()-It wakes up one single thread that called wait() on the same object.

notifyAll()-It wakes up all the threads that called wait() on the same object.

Thread priority is a priority assigned to each thread, which affects the order in which it is scheduled for running.

Threadobject.setPriority(intNumber)

–MIN_PRIORITY = 1

–NORM_PRIORITY=5

–MAX_PRIORITY=10

Try: used to enclose the code that might throw an exception

Catch: used to handle the Exception by declaring the type of exception within the parameter

Finally: used to execute important code such as closing the connection, stream etc. finally block is always executed whether an exception is handled or not.

Throw: This keyword is used to throw an exception for a method, it cannot throw multiple exceptions.

- **throw** is followed by an object (new type)
- used inside the method

Throws: Used to indicate what exception type may be thrown by a method, it can declare multiple exceptions.

- **throws** is followed by a class
- and used with the method signature

Note: For each try block there can be zero or more catch blocks, but only one finally block.

Unit 4

Java swing is used to create window-based applications or desktop applications.(javax.swing)

Steps to create GUI:

Create instance of JFrame----Set size of JFrame---Set layout managers---Create instance of JButton,JTextFieldetc..

Set position and size of a button,TextFieldetc---Add component in Frame---Set Visible True For JFrame
setBounds(intx-coordinate, int y-coordinate, int width, int height)

2 ways to create a frame:-->By creating the object of Frame class -->By extending Frame class

Color c1 = new Color(102, 255, 102);

jf.getContentPane().setBackground(c1);

JLabell=new JLabel(new ImageIcon("bimg2.jpg"));

JButton class is used to create a labelled button which perform some action when the button is pushed. It inherits Abstract Button class.

Steps to perform action on button click

Importing Packages: javax.swing; java.awt; java.awt.event

Create Class which Implementing ActionListener Interface

Create Button and add in Frame:

Register ActionListener to the JButton. For this, we have to call addActionListener() method using the object of the JButtonclass.

Override actionPerformed(ActionEvent e) method

Object of **JLabel class** is a component for placing text in a container. It is used to display a single line of read only text. A user cannot edit it directly. It inherits JComponent class.

Object of a **JTextField class** is a text component that allows editing of a single line text. It inherits JtextComponent class.

JRadioButton class is used to create a radio button. It is used to choose one option from multiple options.

JTextArea class represents a multi-line area that displays text. It is used to edit the text and inherits JComponent class.

JComboBox inherits JComponent class. It shows a popup menu that shows a list and the user can select an option from that specified list.

Steps to perform event in JComboBox

✓Implement ItemListenerinterface

✓Register combobox using addItemListener()

✓ Define `itemStateChanged(ItemEvent e)`

To read item from combo box:

`JComboBoxobj.getSelectedItem();`

JTable class is used to display data in rows and columns wise.

JColorChooser class is used to create a color chooser dialog box so that user can select any color. It inherits `JComponent` class. *******`JColorChooser.showDialog(this,"Select a color",initial_color_name);`

JProgressBar class is used to display the progress status of the task. It inherits `JComponent` class.

JSlider class is used to create slider in GUI. By using `Slider`, we can select a value from given range.

LayoutManager is an interface that is used to arrange components in a specific manner.

Null Layout means that no layout manager is assigned and the components can be put at specific x, y coordinates.

border layout places components in upto five areas: center, north, south, east and west. Each area can contain only one component.

GridLayout class arranges the components in a rectangular grid.

flow layout manager arranges components in a row from left to right, starting a new row if no more components fit into a row. **Default layout for JPanel.**

Box Layout class is used to arrange components either vertically(along Y-axis) or horizontally.

Card Layout class manages the components in such a manner that only one component is visible at a time.

Event Listener is an object which is notified when an event occurs.

Unit 5

JDBC is a software tool and also known as an application programming interface (API) that is used to interact with the database.

Need for Java JDBC

- For establishing stable data base connectivity with the application API.
- To execute SQL queries and DDL/DML commands.
- For viewing and modify data records.

JDBC API: It provides various methods and interfaces for easy communication with the database.

JDBC Driver manager loads a database-specific driver in an application to establish a connection with a database.

JDBC drivers are client-side adapters that convert requests from Java programs to a protocol that the DBMS can understand.

There are 4 types of JDBC drivers:

JDBC-ODBC bridge driver: it provides a bridge to access the ODBC driver installed on each client. Using the JDBC-ODBC bridge driver we can access the databases which support only ODBC.

Native-API driver: uses the client-side libraries of the database. The driver converts JDBC method calls into native calls of the database API. It is not written entirely in java.

Network Protocol driver: uses middleware (application server) that converts JDBC calls directly or indirectly into the vendor-specific data base protocol. It is fully written in java.

Thin driver converts JDBC calls directly into the vendor-specific data base protocol. It is fully written in Java language.

The name of the class that implements `java.sql.Driver` in MySQL Connector/J is `com.mysql.cj.jdbc.Driver`

Steps to Connect Java JDBC

Import-Packages: `java.sql*;`

Load and Register the Driver: `DriverManager.registerDriver(driverClassName);`

Connect to the Database: `getConnection()` method of `DriverManager` class is used to establish connection with the database.

Create Statement: The `createStatement()` method of `Connection` interface is used to create statement. The object of statement is responsible to execute queries with the database.

Execute the query: `executeQuery()` method of `Statement` interface is used to execute queries to the database. This method returns the object of `ResultSet` that can be used to get all the records of a table.

Close the connection object: By closing connection object statement and ResultSet will be closed automatically. The close() method of Connection interface is used to close the connection.

Connection interface represents a session between java application and database. All SQL statements are executed and results are returned within the context of a Connection object.

Statement interface is used to create SQL statements. It provides methods to execute queries with the database.

There are different types of statements that are used in JDBC as follows:

•**Create a Statement:** It is generally used for general-purpose access to databases and is useful while using static SQL statements at runtime.

•**PreparedStatement:** Prepared Statement represents are compiled SQL statement, that can be executed many times. This accepts parameterized SQL queries. In this, “?” is used instead of the parameter.

•**CallableStatement:** The Callable Statement interface is used to execute the SQL stored procedure in a database.

ResultSet Interface: It is used to store the data which are returned from the database table after the execution of the SQL statements. The object of ResultSet maintains cursor point at the result data.

ResultSetMetaData interface provides methods to get metadata from the ResultSet object.

DatabaseMetaData interface provides methods to get metadata of a database such as database product name, database product version, driver name, name of total number of tables, name of total number of views etc.

Unit 6

Networking is a concept of connecting two or more computing devices together so that we can share resources.

Network programming refers to writing programs that execute across multiple devices, in which the devices are all connected to each other using a network.

Port Number is associated with Ip address for communication between two applications.

A media access control address (**MAC address**) is a unique identifier assigned to a network interface controller (NIC) for use as a network address in communications within a network segment.

In connection-oriented protocol, acknowledgement is sent by the receiver. So, it is reliable but slow.

Example is TCP. **In connection-less protocol**, acknowledgement is not sent by the receiver. So, it is not reliable but fast. The example of connection-less protocol is UDP.

A **socket** is an endpoint between two-way communication.

Socketclass objects help us in implementing all fundamental socket operations such as sending, reading data and closing connections.

ServerSocketClass is used for providing system-independent implementation of the server-side of a client/server Socket Connection.

URL class represents an URL. URL known as Uniform Resource Locator. It is simply a string of text (**that contains protocol, host name and file name**) that identifies all the resources on the Internet, telling us the address of the resource, how to communicate with it, and retrieve something from it.

URLConnection class represents a communication link between the URL and the application. It can be used to read and write data to the specified resource referred by the URL.

DatagramSocket class represents a socket for sending and receiving datagram packets where each packet is individually addressed and routed.